

Do moving-average signals deserve our trust?

Golden-cross Buy/Sell analysis of six NSE stocks · January 2015 – July 2018 · SQL (MySQL 8) + Streamlit dashboard

Insights report · Abhijit Sinha

6 NSE stocks	889 trading days each	–50% one-day “crash” that was a bonus issue	+52.4% TCS after adjustment (raw: –23.8%)	0 of 6 stocks where signals beat holding
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Executive summary

- **Two “crashes” were bonus issues.** TCS (2018-05-31) and Infosys (2015-06-15) each fell about 50% in one day because of 1:1 bonus issues, not because value was lost.
- **The fix reverses the winners and losers.** On raw prices TCS is –23.8% and Infosys –30.9%; adjusted, they are **+52.4%** and **+38.2%**, third and fourth best of the six.
- **The price cliff distorts signals.** It created one false Sell for TCS and, for Infosys, hid three real signals and created one false Buy.
- **Following the signal did not beat holding.** On adjusted prices the golden-cross strategy trailed buy-and-hold for all six stocks, with only 6–12 trades per stock, so this is indicative, not proven.
- **Some course-deck figures should not be reused.** Its TCS and Infosys numbers and the TCS “shifting down” conclusion change once the data is checked (page 4).

Problem, objective and stakeholder

Problem	Retail investors follow a <i>golden cross</i> (the 20-day average crossing the 50-day average) without checking whether the signal is reliable or whether the price data is clean.
Objective	Build the signals in SQL, measure how often they fire and how they compare with simply holding, and test whether data problems distort the conclusions.
Stakeholder and decision	A retail investor or junior analyst deciding whether to use a 20/50-day golden-cross signal, and on which stocks.
Success measures	(1) Every checkpoint in the brief is reproduced – 13 of 13 pass. (2) Winners and losers are reported on adjusted prices. (3) Every claim has evidence and a caveat.

Method in one paragraph

This project, **SQL – Stock Market Analysis**, loaded daily closing prices into MySQL. A 20- and 50-day moving average was built with window functions (ROWS BETWEEN 19 / 49 PRECEDING AND CURRENT ROW), left empty until a full window existed. A Buy is the day the 20-day average moves above the 50-day average (yesterday it was at or below); a Sell is the mirror image. All six stocks were handled in one query using PARTITION BY stock. Because two stocks showed implausible one-day drops, prices before each event were divided by the bonus factor of 2, the averages and signals were rebuilt, and the strategy was compared with buy-and-hold.

1. Data and quality checks

Daily NSE prices for Bajaj Auto, Eicher Motors, Hero MotoCorp, Infosys, TCS and TVS Motors – 889 trading days each, 1 Jan 2015 to 31 Jul 2018 – supplied with the course. The source, licence and collection method are not stated in the files, so they should be confirmed with the course provider. Only close_price is analysed.

Check	Result
Rows per stock	889 in every table; identical trading dates across all six; no duplicate dates
Missing values	Only deliverable_qty and pct_deliv_qty: 1 row per stock (0.11%). Eicher, Hero, TCS and TVS on 2015-12-09; Bajaj and Infosys on 2017-08-31.
How handled and why	Left as NULL, rows kept. These columns are not used in any calculation. Gaps on the same two dates across unrelated companies suggest an exchange reporting gap, not a company issue.
Close price	No NULL or zero close prices, so no rows needed repair.
Outliers	Two ~50% one-day drops investigated and explained (section 3); they were corrected, not deleted.

Assumptions (stated as assumptions)

- Only closing prices are used; dividends, brokerage, taxes and market-wide moves are ignored.
- The strategy is long-only and trades at the same-day close the signal is computed on, which is slightly optimistic.
- The ex-bonus dates are the first day the price trades at the new level; prices before that date are divided by the adjustment factor 2.
- 2018 holds only 7 months (146 trading days), so 2018 is never described as a full year.

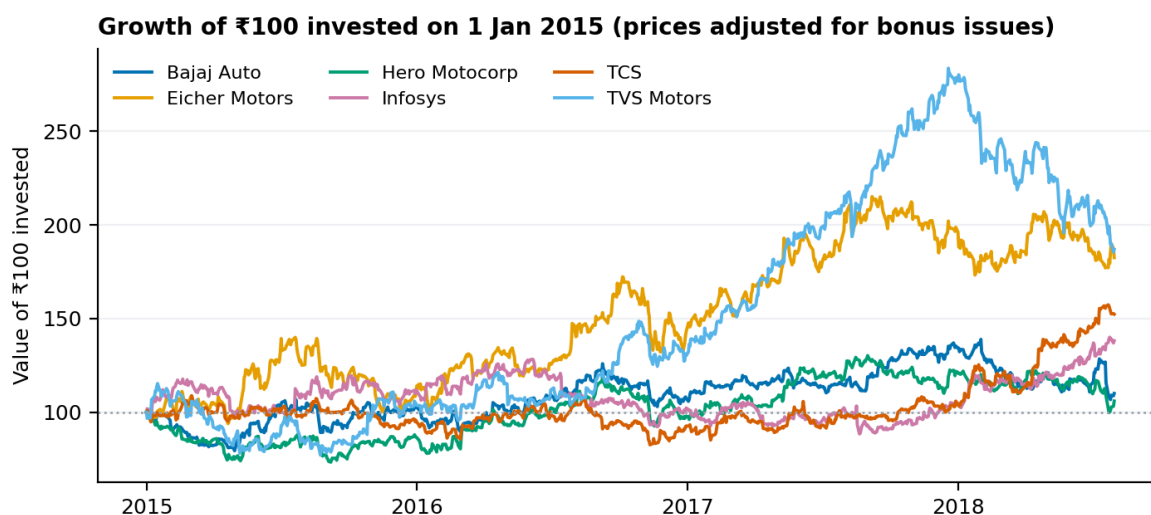
Sampling bias

Six hand-picked large companies (four automakers, two IT firms) over about 3.5 years, covering one market phase. Results cannot be generalised to the whole market.

Sources for the corporate actions

- TCS: NSE circular NSE/FAOP/37878 dated 30 May 2018 – bonus 1:1, ex-date 31 May 2018, record date 2 June 2018, adjustment factor $(1+1)/1 = 2$.
- Infosys: Business Standard / PTI, 15 June 2015 – stock turned ex-bonus 1:1 on 15 June 2015; NSE close Rs 990.45 after adjustment against Rs 1,975.05 before; record date 17 June 2015.
- Infosys also issued 1:1 bonuses on 2 Dec 2014 and 4 Sep 2018 (corporate-action listings); both fall outside the data window, so no further adjustment was needed.

2. How the six stocks performed



Takeaway. TVS Motors and Eicher Motors grew the most (+86.9% and +82.6%). After adjustment TCS (+52.4%) and Infosys (+38.2%) are clear gainers; Bajaj (+10.0%) and Hero (+6.0%) were roughly flat by comparison.

Stock	First close (₹)	Last close (₹)	Change raw	Change adjusted	Buys / Sells raw	Buys / Sells adjusted	Latest signal (adj.)
TVS Motors	276.85	517.45	+86.9%	+86.9%	8 / 8	8 / 8	Sell, 17 May 2018
Eicher Motors	15,239.15	27,820.95	+82.6%	+82.6%	6 / 7	6 / 7	Sell, 6 Jun 2018
TCS	2,548.20	1,941.25	-23.8%	+52.4%	12 / 13	12 / 12	Buy, 20 Apr 2018
Infosys	1,975.80	1,365.00	-30.9%	+38.2%	9 / 9	10 / 10	Buy, 7 May 2018
Bajaj Auto	2,454.10	2,700.70	+10.0%	+10.0%	12 / 11	12 / 11	Buy, 21 Jun 2018
Hero MotoCorp	3,107.30	3,293.80	+6.0%	+6.0%	9 / 9	9 / 9	Sell, 22 May 2018

Totals: 56 Buys and 57 Sells on raw prices (the brief's checkpoint); 57 and 57 on adjusted prices. "Change" compares the first and last trading day, so it ignores everything in between.

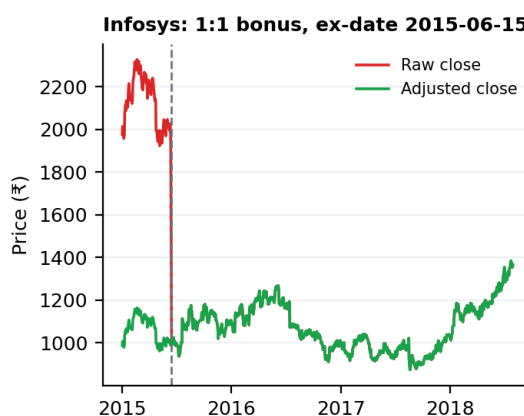
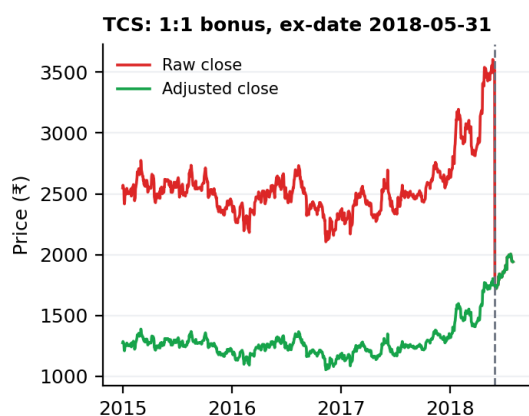
Does the latest signal agree with the long-run trend?

For Bajaj, Infosys and TCS the latest signal is a Buy and the full-period change is positive, so they agree. For Eicher and TVS the latest signal is a Sell although both gained more than 80% over the period, and for Hero it is a Sell with only +6%. A single crossing says the 20-day average has dipped below the 50-day average; it says nothing about the 3.5-year trend. The long-run trend is the better guide for a long-horizon investor, while the signal is useful only as a short-term caution – and with the evidence in section 4, neither should be acted on alone.

3. The data trap: two price cliffs

Each stock's worst one-day move was computed. Four stocks sit between -6.2% and -9.7%; two are about -50%, far outside normal behaviour for large profitable companies.

Stock	Date	Close (₹)	One-day move
TCS	2018-05-31	1,744.80	-50.4%
Infosys	2015-06-15	991.10	-49.9%
TVS Motors	2016-05-03	287.90	-9.7%
Eicher Motors	2015-08-24	18,571.75	-9.3%
Bajaj Auto	2015-08-24	2,188.45	-9.1%
Hero MotoCorp	2018-07-23	3,163.90	-6.2%



Takeaway. Dividing every price before the event by 2 removes the cliff. The two "losers" on raw prices become gainers: TCS +52.4% and Infosys +38.2%.

What the cliff did to the signals

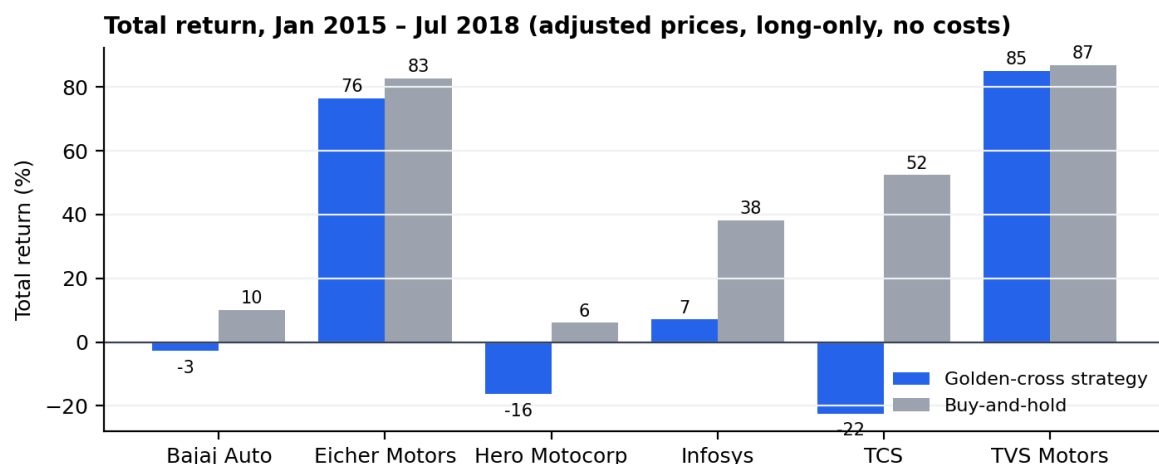
Stock	Signals raw	Signals adjusted	Difference
TCS	12 Buys, 13 Sells	12 Buys, 12 Sells	The Sell on 2018-06-05 exists only on raw prices – a false signal created by the cliff. The other 24 signal days are identical.
Infosys	9 Buys, 9 Sells	10 Buys, 10 Sells	Hidden by the cliff: Buy 2015-07-01, Sell 2015-07-08, Buy 2015-07-27. Created by the cliff: false Buy 2015-08-18.

Which conclusions in the course deck change?

- **Infosys change:** the deck shows –3%; the data gives –30.9% on raw prices and **+38.2%** once adjusted.
- **TCS change:** –23.8% is the correct raw figure, but the price pair shown for TCS in the deck appears mislabelled, and the figure is misleading because of the bonus issue; adjusted it is **+52.4%**.
- **TCS trend:** the deck's last TCS signal (Sell, 5 June 2018, “shifting down”) is the false one; on adjusted prices the latest TCS signal is a **Buy** (20 April 2018).
- **Signal counts:** TCS 12/13 → 12/12; Infosys 9/9 → 10/10. The other four stocks are unchanged.
- **Eicher:** 82.5% in the deck against 82.56% in the data, which rounds to 82.6% (a rounding difference).
- Every number in this report was recomputed from the database; none was copied from the deck unchecked.

4. Did following the signals pay off?

Each Buy was paired with the next Sell on adjusted prices (long-only, in cash otherwise). A final Buy with no Sell is marked to the last close. This extends the brief with a baseline (buy-and-hold) and a stability check – the equivalents, for a SQL project, of a baseline model and an overfitting check.



Takeaway. The strategy trailed buy-and-hold for all six stocks. It came closest for the two strongest trending stocks, Eicher and TVS, and was furthest behind for TCS, where it missed most of the rise.

Stock	Trades (n)	Win rate	Round trips ≤ 30 days	...of which lost money	Strategy	Buy-and-hold
Bajaj Auto	12	33%	5	4	–2.7%	+10.0%
Eicher Motors	6	100%	0	0	+76.5%	+82.6%
Hero MotoCorp	9	33%	1	1	–16.3%	+6.0%
Infosys	10	40%	2	2	+7.1%	+38.2%
TCS	12	25%	1	1	–22.4%	+52.4%
TVS Motors	8	38%	2	2	+85.0%	+86.9%

Open trades at the end (marked to the last close): Bajaj Buy of 21 Jun 2018 (–4.7%), Infosys Buy of 7 May 2018 (+15.4%), TCS Buy of 20 Apr 2018 (+14.0%). No costs, dividends or taxes are included.

Whipsaws: signals flipping within a few weeks

Eleven round trips lasted 30 days or less and ten of them lost money. Bajaj Auto had five (four losing, for example a -9.6% trade from 28 Dec 2015 to 18 Jan 2016 and -8.0% in early 2018). TVS Motors lost 11.0% on a 23-day trade in April–May 2018. Following every signal therefore means paying for the frequent false starts, before any brokerage cost.

Is it stable over time?

Average return per trade changed sign or size between the 2015–16 and 2017–18 entry periods for several stocks: TCS $-4.96\% \rightarrow +0.26\%$, Infosys $-1.07\% \rightarrow +3.48\%$, TVS $+2.29\% \rightarrow +37.86\%$ (the latter rests on just 2 trades), Hero $-0.57\% \rightarrow -2.64\%$. The signal is not consistently reliable, which is a warning against overfitting to one period.

A moving average only uses past prices. What does that mean for how early a golden cross can tell you anything?

The 50-day average cannot exist before 50 trading days of data, so no signal is possible before mid-March 2015. After that, a crossing confirms that the recent average has already overtaken the longer one – the move is partly done. The signal lags the trend by construction; it cannot predict a turn.

5. Claims, evidence and caveats

Claim	Evidence (query / number)	Caveat
TCS and Infosys are not losers; the cliffs are bonus issues.	Task 12: -50.4% (2018-05-31) and -49.9% (2015-06-15). Task 13: $+52.4\%$ and $+38.2\%$. NSE circular FAOP/37878; Business Standard, 15 Jun 2015.	Dividends ignored; adjustment assumes the 1:1 ratio is exact for the whole window.
The price cliff creates and hides signals.	Raw vs adjusted signal comparison: TCS 1 difference, Infosys 4.	Only the 20/50 window pair was tested.
Winners after adjustment: TVS, Eicher, TCS, Infosys; Bajaj and Hero are roughly flat.	Task 11 and 13 changes (first vs last day).	First-to-last-day change ignores the path; it depends on the start and end dates chosen.
Golden-cross trading did not beat buy-and-hold on any stock.	Strategy vs buy-and-hold table (section 4).	6–12 trades per stock, no costs or dividends, same-day-close execution, one 3.5-year window.
Signals flip frequently, and many quick flips lose money.	10 of 11 round trips ≤ 30 days lost money.	Small counts; the sample cannot support a statistical test.
The latest signal is context, not a forecast.	Latest signals vs long-run change (section 2).	Single crossing; depends on window choice.
Missing data does not affect the results.	1 NULL per stock in two unused columns.	Rows are kept; if those columns are used later the gaps must be handled.

6. What is not in this data that a real investor would need

- Dividends and other corporate actions (only the two bonus issues were found and adjusted).
- Brokerage, taxes and market impact – they would lower the strategy returns further.
- News, results announcements and sector events that drive price changes.
- The broader market (index returns) for a fair benchmark, and risk measures such as drawdown.
- Out-of-sample data: every figure here is in-sample for one 3.5-year window.

7. Recommendations: from finding to action

Each recommendation follows a reasoning chain – finding → why it matters → action → how it is measured – and is written for a specific investor type. They are **ideas to test**, not proven conclusions: six stocks and 6–12 trades each.

For	1 · Finding	2 · Why it matters	3 · Action	4 · How to measure
Data analyst affects 2 of 6 stocks	TCS and Infosys fell ~50% in one day (bonus issues): raw –23.8% / –30.9%, adjusted +52.4% / +38.2%.	Raw data reversed winners and losers and created or hid signals, so later numbers were wrong.	Keep exchange-confirmed events in corporate_events; rebuild from it; flag moves above 30%.	No unexplained one-day move above 30%; events match exchange notices.
Long-term holder affects 6 of 6 stocks	The signal strategy trailed buy-and-hold for 6 of 6 stocks; on ₹1,00,000 per stock the shortfall was about ₹1.6 lakh after 0.1% costs.	Averages lag: exits come after part of a fall and re-entries after part of a rise.	Hold through signals; treat a Sell as a prompt to review, not an instruction.	A rule is adopted only if it beats holding after costs on unseen data.
Short-term trader 5 of 6 stocks had quick flips	10 of 11 round trips of 30 days or less lost money.	Frequent flips pay costs and losses repeatedly; only strong trenders (Eicher, TVS) came close to holding.	Test ignoring a signal within 30 days of the previous one; try 10/30 and 50/200.	Fewer losing quick trips and a result that beats holding in both 2015–16 and 2017–18.
Any investor stocks now showing Sell	Eicher (+82.6%) and TVS (+86.9%) show a latest Sell after large gains.	One crossing says nothing about a multi-year trend.	Treat the signal as a caution flag; record the reason for any decision.	Decisions logged with signal and long-run change; reviewed after 3 and 6 months.

Impact in rupees (a stated assumption, not a forecast)

If ₹1,00,000 had been put into each stock on 1 Jan 2015 (adjusted prices, long-only, 0.10% cost on every buy and sell, no tax or dividends):

Stock	Buy-and-hold	Follow the signals	Difference
Bajaj Auto	₹1,10,000	₹95,087	–₹14,913
Eicher Motors	₹1,82,600	₹1,74,394	–₹8,206
Hero MotoCorp	₹1,06,000	₹82,206	–₹23,794
Infosys	₹1,38,200	₹1,05,083	–₹33,117
TCS	₹1,52,400	₹75,835	–₹76,565
TVS Motors	₹1,86,900	₹1,82,062	–₹4,838
All six	₹8,76,100	₹7,14,666	–₹1,61,434

Try your own amount, stock, investor type and cost on the dashboard's Recommendations page.

Limitations and biases

- **Selection / survivorship bias:** six hand-picked large companies (four autos, two IT) that survived and did well.
- **One market phase, in-sample:** 3.5 years that were mostly rising; every result is measured on the data it was built on.
- **Execution (look-ahead) bias:** trades assumed at the same-day close the signal is computed on, which flatters the strategy.
- **Small samples:** 6–12 trades per stock, so differences between stocks may be chance; no significance test was possible.
- **Causality:** everything is an association over history; nothing shows that a signal causes a return.
- **Left out:** taxes, dividends, news and costs beyond the illustration; only two bonus issues adjusted; data source and licence undocumented.

If new data arrives (distribution shift)

Later data will contain new bonus or split events (Infosys, for example, issued another 1:1 bonus in September 2018). Add each event to `corporate_events` and re-run `sql/06` and `sql/07`; no other change is needed. Signals and averages are recomputed from scratch, so nothing stale is carried over.

Monitoring and refresh plan

Load new prices monthly; run `scripts/validate_checkpoints.py` and a check that flags any one-day move beyond about 30% for manual review before the dashboard tables are rebuilt.

Three next steps

- Test other windows with a hold-out period.
- Add transaction costs and dividends and execute at the next day's open.
- Automate the refresh and the price-cliff check.

8. Acknowledgement

I thank **Labmentix** for the project brief, the student guide and the dataset, and my mentors and instructors for their guidance and feedback. Thanks also to the **National Stock Exchange of India** (public corporate-action circular) and **Business Standard / PTI**, whose published reports confirmed the bonus-issue dates; to **Aiven** (cloud MySQL) and **Streamlit** (app hosting); and to the open-source tools behind this work – MySQL, Python, pandas, SQLAlchemy, Plotly, Matplotlib and ReportLab. I also used the **Claude AI assistant** for workflow planning, code review and drafting support; every number in this report was re-checked against my own queries. Company logos shown in the dashboard are trademarks of their respective owners and are used for identification only.

Thank you

Thank you for reading this report. I welcome your feedback and questions.

Live dashboard: nse-stock-signals-sql-dashboard-jkxkfdhnxtrswkl6beakt.streamlit.app

Code: github.com/abhi-1009/nse-stock-signals-sql-dashboard

Learning project – not investment advice.